NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

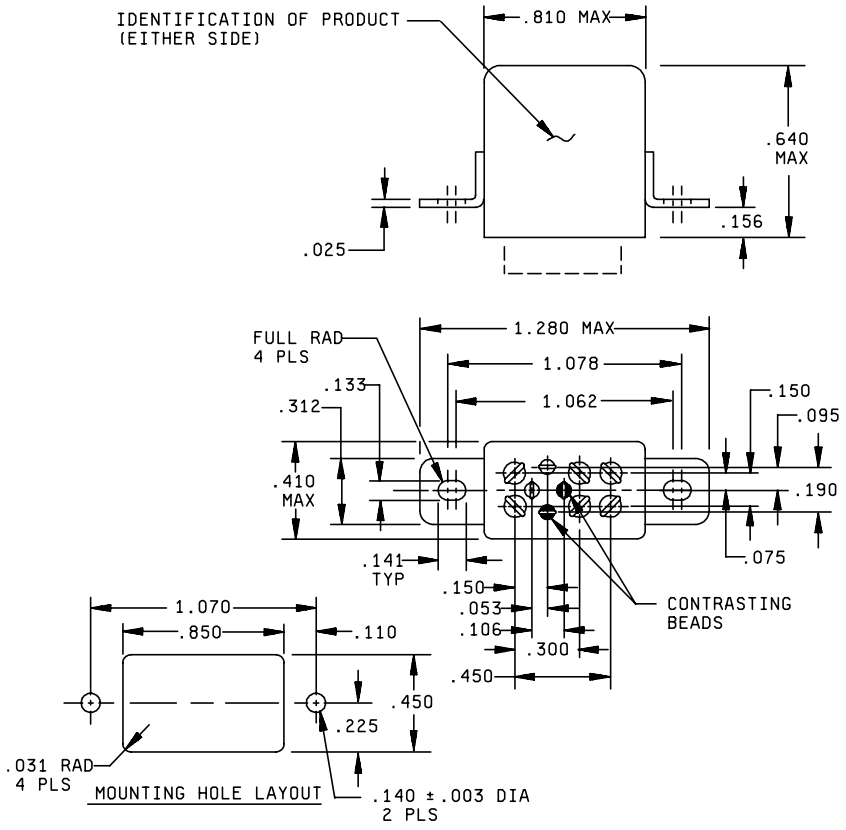
MIL-PRF-6106/38B(USAF)
15 September 1993
SUPERSEDING
MIL-R-6106/38A(USAF)
7 September 1984

PERFORMANCE SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, TYPE I, MAGNETIC LATCH, PERMANENT MAGNET DRIVE, LOW LEVEL TO 5 AMPERES, 2PDT, HERMETICALLY SEALED

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification list in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-PRF-6106.

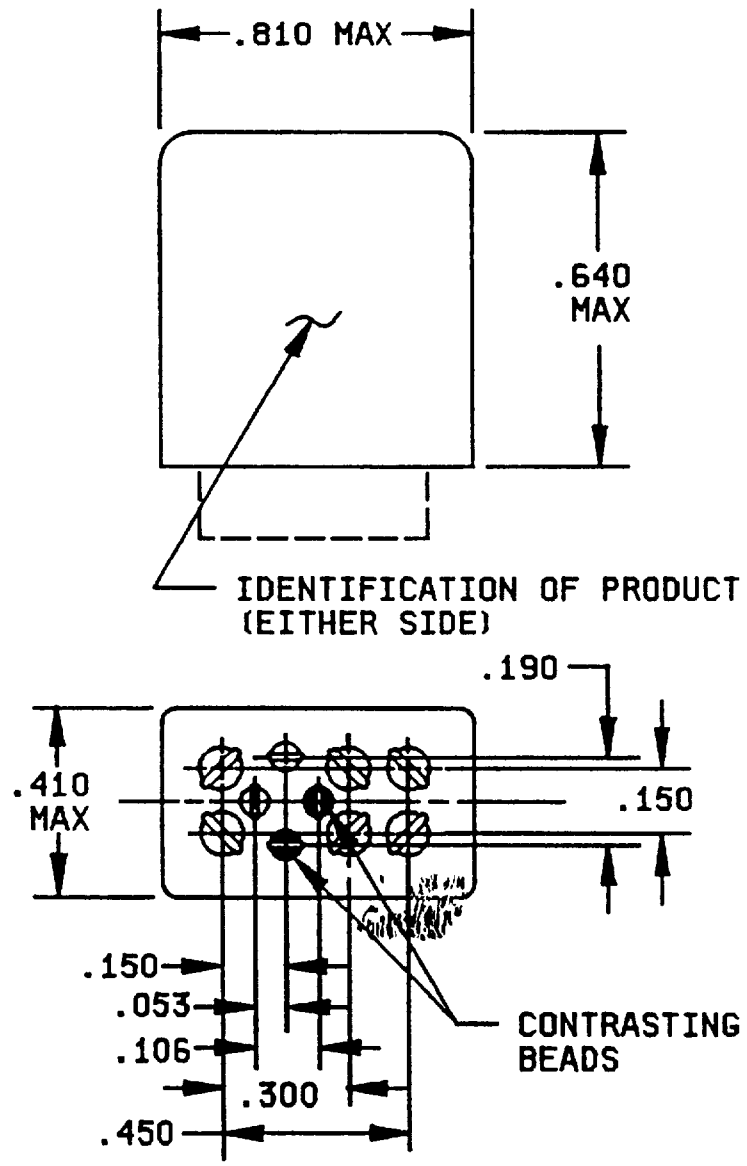


Configuration A

FIGURE 1. Outline drawing.

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

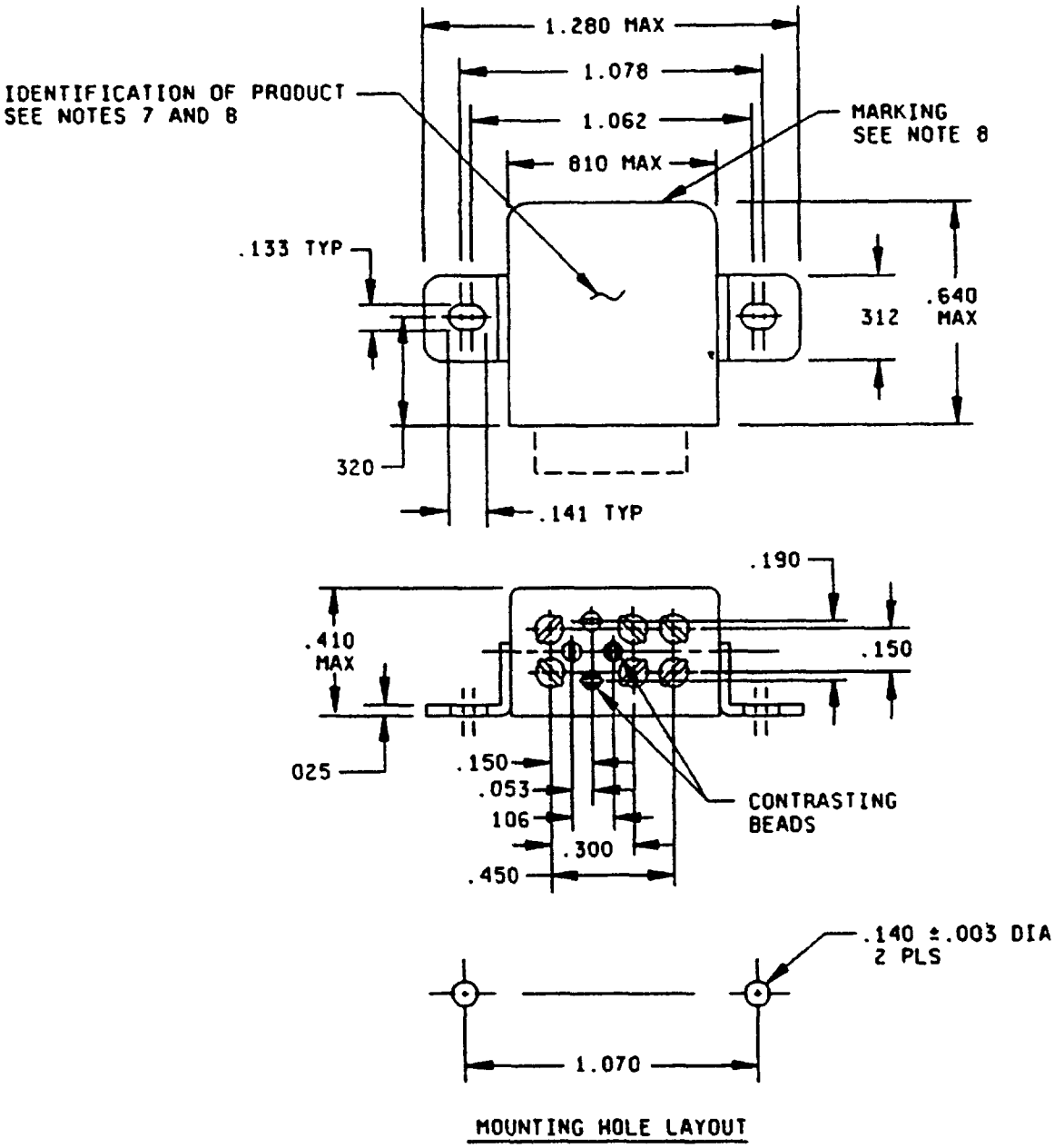
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Configuration B

FIGURE 1 Outline drawing - Continued.

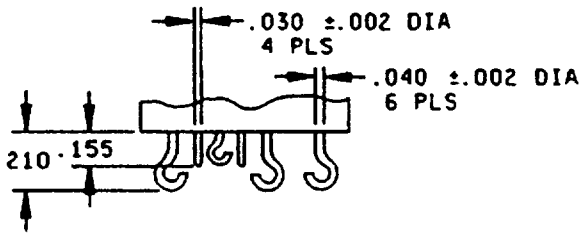
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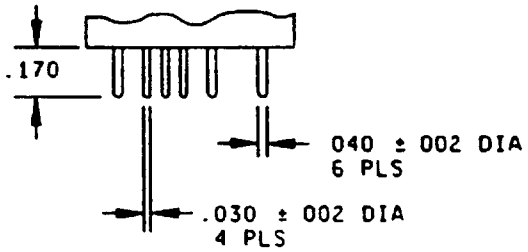
Configuration C

FIGURE 1. Outline drawing - Continued

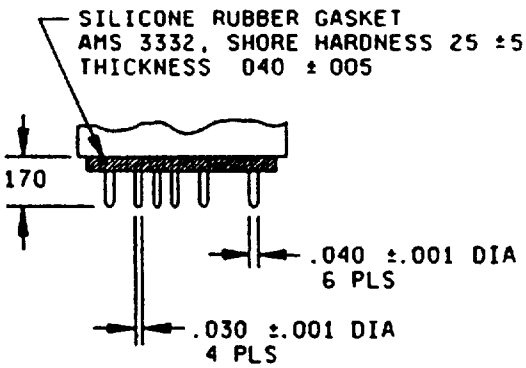
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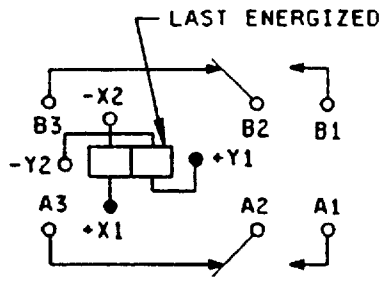
SOLDER HOOK
(TIN PLATED)



SOLDER PIN
(TIN PLATED)



SOCKET PIN
SEE NOTE 9



CIRCUIT DIAGRAM

TO CLOSE NUMBER 1 CONTACTS (OPERATE), ENERGIZE X1 AND X2.

TO CLOSE NUMBER 3 CONTACTS (RESET), ENERGIZE Y1 AND Y2

FIGURE 1 Outline drawing - Continued

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Inches	mm	Inches	mm	Inches	mm
001	0.03	.106	2.69	.300	7.62
.002	0 05	.133	3.38	312	7 92
.003	0.08	.140	3 56	.410	10 41
005	0 13	.141	3.58	.450	11 43
025	0 64	.150	3.81	640	16 26
030	0 76	.155	3 93	.810	20 57
031	0 79	156	3 96	850	21 59
040	1 02	170	4 32	1 062	26 97
053	1 34	190	4 82	1 070	27 18
075	1 90	.210	5.33	1 078	27 38
095	2 52	.225	5 72	1 280	32 51

NOTES

1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 inch (25 4 mm)
4. There shall be affixed to the relay a legible circuit diagram that identifies each terminal location specified.
5. These relays are polarized bistable
6. Permanent magnet drive consists of a permanent magnet with its flux path switched and combined with the electromagnetic flux.
7. Applicable to configuration C only The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side
8. Applicable to configuration C only. Relays shall be marked with the manufacturer's name or source code and date code In addition, ER relays shall be marked with the serial number Marking shall appear with the bottom of the print adjacent to the near side
9. Socket terminals shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick, minimum

FIGURE 1. Outline drawing - Continued

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REQUIREMENTS

CONTACT DATA·

Life Unless otherwise specified 100,000 cycles

Operational reliability. Applicable, the alternate low level method shall be used 1/ 2/ 3/

Load ratings·

High level (relay case grounded)

Resistive 5 amperes at 28 V dc, 115 V ac, 400 Hz, 1 phase, 115/200 V ac, 400 Hz, 3 phase

Inductive 3 amperes at 28 V dc, 5 amperes at 115 V ac, 400 Hz, 1 phase, 115 V ac, 400 Hz, 3 phase (Life 20,000 cycles)

Motor 2 amperes at 28 V dc, 3 amperes at 115 V ac, 400 Hz, 1 phase, 115 V ac, 400 Hz, 3 phase

Lamp 1 ampere at 28 V dc

Low level: 10 to 50 µA at 10 to 50 mV dc or peak ac 1/ 2/ 3/

Intermediate current Applicable

Transfer load Not applicable

Mechanical life (reduced current) 1.25 amperes at 28 V dc, 115 V ac, 1 phase and 3 phase (Life· 400,000 cycles)

Mixed loads· Applicable 1/

Contact voltage drop or resistance 2/ 3/

High level
Initial. 0 150 volt maximum
After life 0 175 volt maximum

Low level:
Initial: 0 050 ohm maximum
After life 0 150 ohm maximum

Contact bounce· 0.5 millisecond maximum

1/ Relay shall be capable of switching low level while switching any of the other rated loads on adjacent contacts

2 For low level, the following shall apply·
a During endurance and operational reliability testing, contact load shall be 10 to 50 microamperes, 10 to 50 millivolts open circuit voltage, 100 ohms maximum contact resistance
b Static contact resistance shall be performed at 50 mA maximum, 50 mV maximum

3/ For group A contact voltage drop test, high level testing shall be performed first, followed by low level testing The contacts shall not make or break the high level load

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Overload current 20 amperes dc
30 amperes ac

Rupture current 25 amperes dc
40 amperes ac

Time current relay characteristics. Applicable (see table I)

TABLE I. Time current relay characteristics. 1/ 2/

1	10 A	1 hour
2	15 A	20 seconds
3	37 5 A	2 seconds
4	75 A	53 second

1/ CAUTION Compare with time current characteristics of the associated circuit protective device

2/ All relays shall withstand overload and fault currents Relays must be able to sustain five applications (make and carry only) of power concurrently on adjacent poles at each of four different current levels in the sequence listed in table I Separate relays shall be tested at 28 V dc and 115/200 V ac, 400 Hz, 3 phase Cooling time between successive applications shall be 30 minutes Tests shall be performed on both normally open and normally closed contacts There shall be no failures or evidence of welding or sticking and relays shall pass contact voltage drop at the conclusion

COIL DATA 4/: (See table II)

Operate time 4 milliseconds maximum with rated coil voltage

Duty rating Continuous

ELECTRICAL DATA:

Insulation resistance, initial· 100 megohms.

After life or environmental tests 50 megohms

Dielectric withstanding voltage (sea level):	Initial V rms (60 Hz)	After life tests V rms (60 Hz)
Coil to case	500	350
Coil to coil	500	350
All other points	1,000	750

Dielectric withstanding voltage (altitude) 5/	80,000 ft V rms (60 Hz)	300,000 ft V rms (60 Hz)
Coil to case	200	375
Coil to coil	200	375
All other points	250	500

4/ CAUTION Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations
a Row to row assisting fields 125 inch
b Row to row opposing fields 1875 inch
c Side to side alternating fields .0625 inch
d Side to side like fields 125 inch

5/ Dielectric may be improved by suitable insulation of terminals and wiring after installation

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TABLE II Dash numbers and characteristics 1/

Military PIN M6106/38-	Terminal type	Mounting configuration	Coil data						
			Rated voltage (V dc) 2/	Resistance Ω ±10% at 25°C	Maximum Voltage (V dc) 3/	Maximum Current A	Pickup volt (max) (V dc) Normal 4/	Pickup volt (max) (V dc) High temp test	Pickup volt (max) (V dc) Continuing current test
001	Solder hook	A	28	730	29	044	18	19 8	22 5
002	Socket pin	A	"	"	"	"	"	"	"
003	Solder pin	A	"	"	"	"	"	"	"
004	Solder hook	C	"	"	"	"	"	"	"
005	Solder pin	B	"	"	"	"	"	"	"
006	Solder pin	C	"	"	"	"	"	"	"

1/ Each relay possesses high level and low level capabilities However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications

2/ CAUTION Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.

3/ When maximum ambient temperature does not exceed +85°C, the maximum coil voltage shall be 32 V dc.

4/ Over the temperature range

ENVIRONMENTAL CHARACTERISTICS

Temperature range -70°C to +125°C

Maximum altitude rating 300,000 feet

Shock (specified pulse): MIL-STD-202, method 213, test condition C, except peak value shall be 200 g's Contact chatter shall not exceed 10 microseconds for closed contacts and 1 microsecond maximum for open contacts.

Vibration (sinusoidal): Applicable, 30 g's maximum and frequency range shall be 70 to 3,000 Hz Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum for open contacts

Vibration (random) MIL-STD-202, method 214, test condition IG, (0 $4G^2$/Hz 50-2,000 Hz), 15 minutes each plane.

Acceleration Applicable (15 g's maximum)

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PHYSICAL DATA

Terminal strength

Solder hook terminals

Twist test: Not applicable

Pull force (0 030 diameter) 3 pounds ±0 3 pound

Pull force (0.040 diameter) 5 pounds ±0 5 pound

Solder pin terminals

Bend test. Applicable.

Pull force (0 030 diameter) 3 pounds ±0 3 pound

Pull force (0 040 diameter) 5 pounds ±0 5 pound

Plug-in terminals

Bend test Applicable

Pull force (0 030 diameter): 3 pounds ±0 3 pound

Pull force (0.040 diameter). 5 pounds ±0.5 pound.

Terminal solderability Applicable to solder pin and solder hook terminals only

Dimensions and configurations: See figure 1

Weight (maximum) .036 pound (16 34 grams)

Seal: Hermetic, relays shall be sealed by welding (laser, TIG (tungsten inert gas) or other suitable means as approved by the qualifying activity)

Construction (internal and external) All welded, except that coil magnet wire to coil lead wire shall be soldered

Identification of product. Applicable

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Qualification by similarity: See table III

TABLE III Qualification by similarity

PIN M6106/38-	Basic qualification A (DC)	Dynamics Socket/pins A 1/	B	C	D	Environmental A (DC)
001	4		Config A 2/			4
002	4	Config A 2/				4 2/
003	4		Config A 2/			4
004	4			Config B 2/		4
005	4				Config C 2/	4
006	4			Config B 2/		4

1/ Terminals: Socket pin type terminals are considered worst case Solder pins and hooks may be qualified by similarity

2/ Socket pins (with gasket seals), are considered nonsimilar to solder pins and solder hooks Each type must be tested

Marginal notations are not used on this revision to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodian
Air Force - 85

Review activities
Air Force - 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5945-F736-07)

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